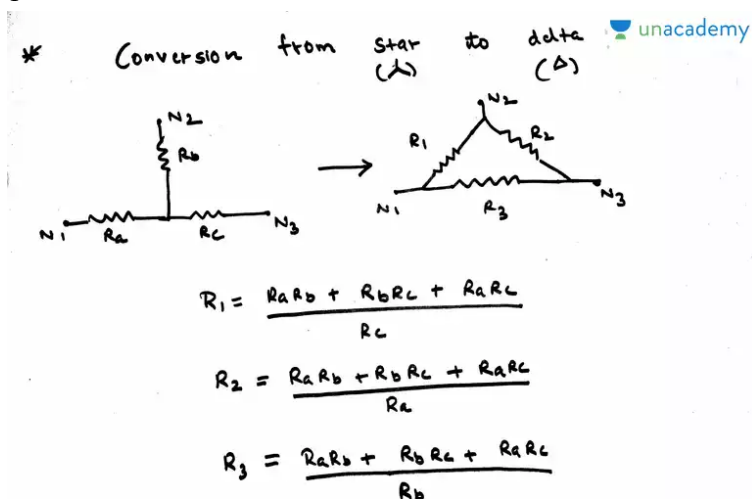


Scheme of Valuation/Answer Key (Scheme of evaluation (marks in brackets) and answers of problems/key)		
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY SECOND SEMESTER B.TECH DEGREE EXAMINATION(2019 SCHEME), JUNE 2022		
Course Code: EST 130		
Course Name Basic Electrical		
Max. Marks: 50		Duration: 1.5 Hours

1



Resistance values are 20Ω , 30Ω , 50Ω

2 Two coils A and B of 500 and 750 turns respectively are connected in series on the same magnetic circuit of reluctance $1.55 \times 10^6 \text{ AT/Wb}$

$L_A = 0.16\text{H}$, $L_B = 0.36 \text{ H}$ and ii) $M = 0.24\text{H}$

3 Explain the concept of statically induced emf in a magnetic circuit

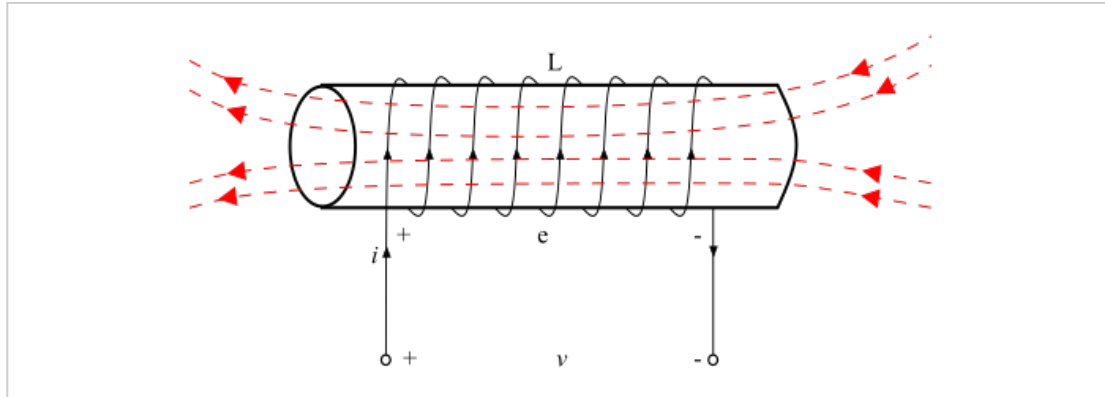
When the conductor is stationary and the magnetic field is changing, the induced EMF in such a way is known as statically induced EMF (as in a transformer).

It is so called because the EMF is induced in a conductor which is stationary.

- Statically Induced EMF
- Mutually Induced EMF

Self-Induced EMF

When an EMF is induced in the coil due to the change of its own magnetic flux linked with it is known as *self-induced EMF*.



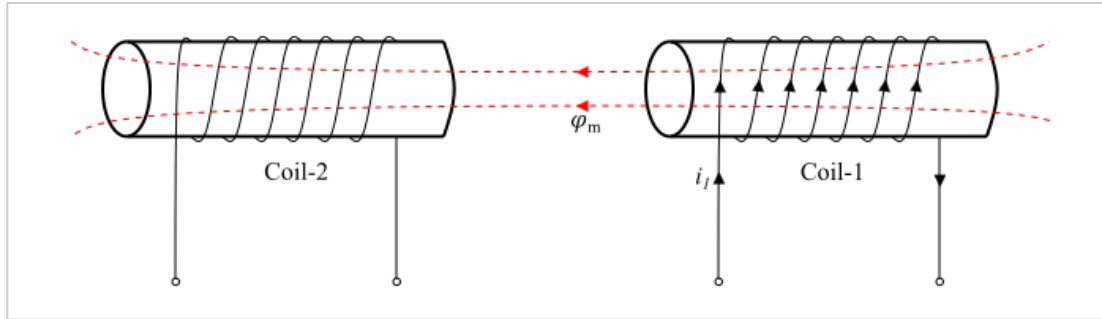
Explanation – When a current flows in a coil, a magnetic field produced by this current through the coil. If the current in the coil changes, then the magnetic field linking the coil also changes. Therefore, according to Faraday’s law of electromagnetic induction, an EMF being induced in the coil. The induced EMF in such a way is known as *self-induced EMF*.

Mathematically, self-induced EMF is given by,

$$e = L \frac{di}{dt} \quad \dots (1)$$

Mutually Induced EMF

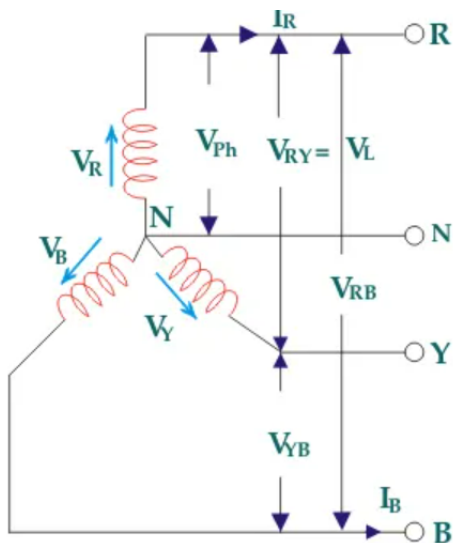
When an EMF is induced in a coil due to changing magnetic flux of neighbouring coil is known as *mutually induced EMF*.



Explanation – Consider two coils coil-1 and coil-2 placed adjacent to each other (see the figure). A fraction of the magnetic flux produced by coil-1 links with the coil-2. This magnetic flux which is common to both the coils 1 and 2 is known as *mutual flux*

(ϕ_m). Now, if the current in coil-1 changes, the mutual flux also changes and thus EMF being induced in both the coils. The EMF induced in coil-2 is known as *mutually induced EMF*, since it is induced due changing in flux which is produced by coil-1.

4



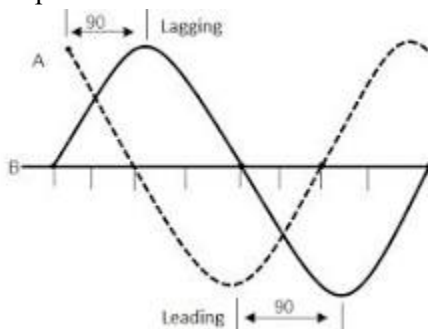
$$\begin{aligned}
 V_L &= |V_{RY}| = \sqrt{V_R^2 + V_Y^2 + 2V_R V_Y \cos 60^\circ} \\
 &= \sqrt{V_{ph}^2 + V_{ph}^2 + 2V_{ph} V_{ph} \times \frac{1}{2}} \\
 &= \sqrt{3} V_{ph} \\
 \therefore V_L &= \sqrt{3} V_{ph}
 \end{aligned}$$

5 Phase -

The phase of an alternating quantity is defined as the divisional part of a cycle through which the quantity moves forward from a selected origin. When the two quantities have the same frequency, and their maximum and minimum point achieve at the same point, then the quantities are said to have in the same phase.

Phase difference - **The time interval by which a wave leads by or lags by another wave is called "Phase difference"**

example:



6 Mesh equations

$$\text{Mesh 1 : } -10i_1 - 20(i_1 - i_2) - 10 = 0$$

$$\text{Mesh 2 : } 40 - 20i_2 + 10 - 10(i_2 - i_3) - 20(i_2 - i_1) = 0$$

$$\text{Mesh 3 : } -10i_3 + 50 - 10(i_3 - i_2) - 10 = 0$$

$$i_1 = 1A ;$$

$$i_2 = 2A ;$$

$$i_3 = 3A$$

7

$$\frac{v_1 - 3}{2} + \frac{v_1}{3} + \frac{v_1 - v_2}{4} = 0$$

$$\frac{v_2 - 2}{1} + \frac{v_2}{2} + \frac{v_2 - v_1}{4} = 0$$

$$v_1 = 1.704V$$

$$v_2 = 1.386V$$

Power Dissipated = 0.025 Watt

8 a) core forms a closed magnetic loop of path length 32 cm. Half of this path has a cross-sectional area of 2 cm² and relative permeability 800. The other half has a cross-sectional area of 4 cm² and relative permeability 400. Find the current needed to produce a flux of 0.4 Wb in the core if it is wound with 1000 turns of insulated wire. Ignore leakage and fringing effects

Reluctance, $S = 1591549.43 \text{ AT/wb}$, $I = 636.8 \text{ A}$

b)

<i>Electric Circuit</i>	<i>Magnetic Circuit</i>
1. E.M.F. in volts	1. M.M.F. in Ampere Turns
2. Current in Amperes	2. Flux in Webers
3. Resistance in Ohms	3. Reluctance in AT/Wb
$R = \frac{\rho l}{a}$	$S = \frac{l}{\mu a}$
4. Conductivity = $\left(\frac{1}{\text{resistivity}} \right)$	4. Permeability = $\left(\frac{1}{\text{reluctivity}} \right)$
5. Conductance = $\left(\frac{1}{\text{resistance}} \right)$	5. Permeance = $\left(\frac{1}{\text{reluctance}} \right)$
6. Current = $\left(\frac{\text{e.m.f.}}{\text{resistance}} \right)$	6. Flux = $\left(\frac{\text{M.M.F.}}{\text{reluctance}} \right)$
7. Resistivity	7. Reluctivity

9a)

Flux density, $B = \frac{\phi}{a} = \frac{0.1 \times 10^{-3}}{6 \times 10^{-4}} = 0.167 \text{ Wb/m}^2$

Ampere-turns required for iron will be :

$$AT_i = H_i \times l_i$$

$$= \frac{B}{\mu_0 \mu_r} \times l_i = \frac{0.167}{4\pi \times 10^{-7} \times 470} \times 0.3 = 84.83 \text{ AT}$$

Ampere-turns required for air will be :

$$AT_g = \frac{B}{\mu_0} \times l_g = \frac{0.167}{4\pi \times 10^{-7}} \times (2 \times 10^{-3}) = 265.8 \text{ AT}$$

$\therefore$ Total AT = $265.8 + 84.83 = 350.63 \text{ AT}$

$\therefore$ Magnetising current, $I = 350.63/\text{N} = 350.63/100 = 3.51 \text{ A}$

b)Relative Permeability

Relative permeability is a dimensionless quantity. It is the ratio of two quantities with the same units, so relative permeability has no unit. Relative permeability of free space is 1.

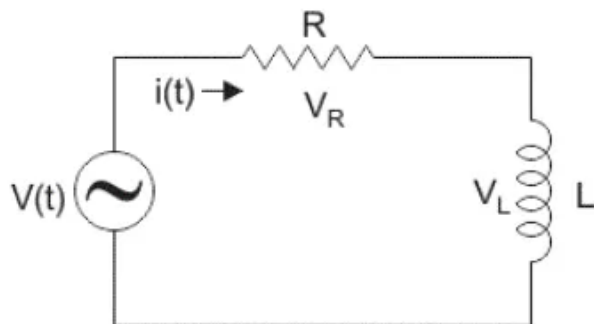
Its expression is given as;

$$\mu_r = \mu / \mu_m$$

Relative permeability = (number of lines of magnetic induction per unit area in a material)/(number of lines per unit area in a vacuum)

The magnetic flux density or magnetic induction is the number of lines of force passing through a unit area of material, B. The unit of magnetic induction is the tesla (T)

10



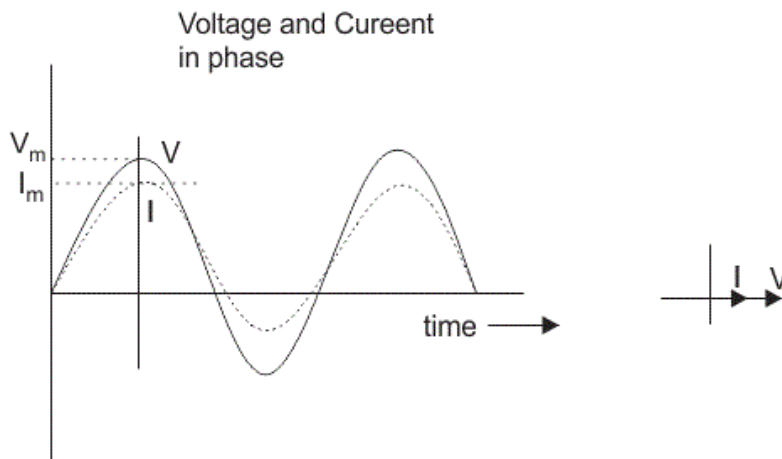
An RL circuit (also known as an RL filter or RL network) is defined as an electrical circuit consisting of the passive circuit elements of a resistor (R)

and an inductor (L) connected together, driven by a voltage source or current source.

Due to the presence of a resistor in the ideal form of the circuit, an RL circuit will consume energy, akin to an RC circuit or RLC circuit.

This is unlike the ideal form of an LC circuit, which will consume no energy due to the absence of a resistor. Although this is only in the ideal form of the circuit, and in practice, even an LC circuit will consume some energy because of the non-zero resistance of the components and connecting wires.

Phasor Diagram



Step- I. In case of series RL circuit, resistor and inductor are connected in series, so current flowing in both the elements are same i.e $I_R = I_L = I$. So, take current phasor as reference and draw it on horizontal axis as shown in diagram.

Step- II. In case of resistor, both voltage and current are in same phase. So draw the voltage phasor, V_R along same axis or direction as that of current phasor. i.e V_R is in phase with I .

Step- III. We know that in inductor, voltage leads current by 90° , so draw V_L (voltage drop across inductor) perpendicular to current phasor.

Step- IV. Now we have two voltages V_R and V_L . Draw the resultant vector (V_G) of these two voltages. Such as, and from right angle triangle we get, phase angle

Power in an RL Circuit

In series RL circuit, some energy is dissipated by the resistor and some energy is alternately stored and returned by the inductor-

1. The instantaneous power delivered by voltage source V is $P = VI$ (watts).
2. Power dissipated by the resistor in the form of heat, $P = I^2R$ (watts).
3. The rate at which energy is stored in inductor,

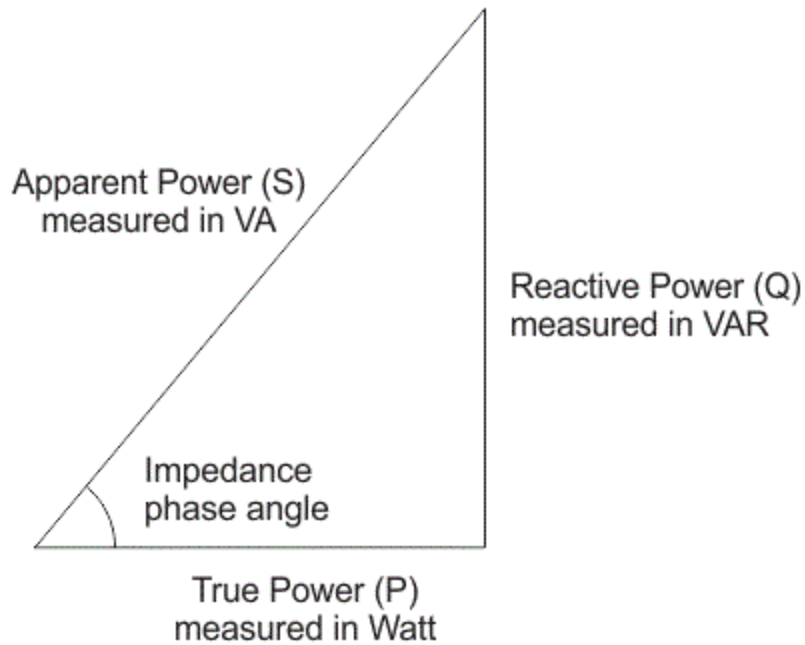
$$P = V_L I = LI \frac{dI}{dt} \text{ (watts)}$$

So, total power in series RL circuit is given by adding the power dissipated by the resistor and the power absorbed by the inductor.

$$P = I^2 R + LI \frac{dI}{dt}$$

Power triangle for series RL circuit is shown below,

The "Power Triangle"



11

Star

$$1) I_{ph} = \frac{V_{ph}}{Z_{ph}} = 3.96 \text{ A}$$

$$2) I_{ph} = I_L = 3.96 \text{ A}$$

$$3) P = \sqrt{3} V_L I_L \cos \phi = 1.412 \text{ kw}$$

Delta

$$1) I_{ph} = \frac{V_{ph}}{Z_{ph}} = 6.86 \text{ A}$$

$$2) \sqrt{3} I_{ph} = I_L = 11.88 \text{ A}$$

$$3) P = \sqrt{3} V_L I_L \cos \phi = 4.23 \text{ kw}$$